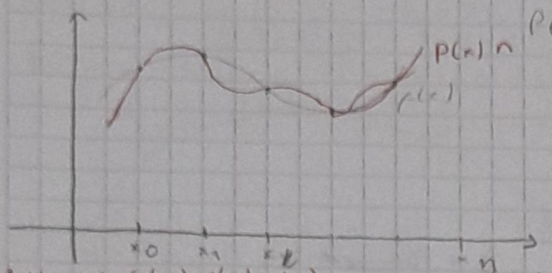
Ders 6



$P'(x_k) = f'(x_k)$

$P(x_k) = f(x_k)$

Hermite Polinomları

$\rightarrow H_{2n+1}(x) = a_0 + a_1 x + \dots + a_{2n+1} x^{2n+1}$

(*) $H(x_k) \Rightarrow f(x_0)\ f(x_1)\ f(x_k)\ \dots\ f(x_n)$ $n+1$

(**) $H'(x_k) \Rightarrow f'(x_0)\ f'(x_1)\ f'(x_k)\ \dots\ f'(x_n)$ $n+1$

(1) $\Rightarrow 2n+2$

Teorem 1: $f \in C^1[a,b]$

$x_0, x_1, \dots, x_n \in [a,b]$

(1) tablosundan tek türlü bir $H_{2n+1}(x)$ belirlenir ve şeyle yazılır

$$H_{2n+1}(x) = \sum_{k=0}^{n} H_{n,k}(x) f(x_k) + \sum_{k} \hat{H}_{n,k}(x) f'(x_k)$$

$$H_{n,k}(x) = [1 - 2(x - x_k) L'_{n,k}(x_k)]\, L^2_{n,k}(x)$$

$$\hat{H}_{n,k}(x) = (x - x_k) L^2_{n,k}(x)$$

$$f(x) = H_{2n+1}(x) + \frac{f^{(2n+2)}(\xi)}{(2n+2)!} (x-x_0)^2 (x-x_1)^2 \dots (x-x_n)^2$$

Örnek 1 $f(x) = \sin(\pi x)$ $x_0 = 0$; $x_1 = \frac{1}{2}$ düğüm noktaları

$f(x_0) = 0$, $f(x_1) = \sin(\pi/2) = 1$

$f'(x_0) = \pi$, $f'(x_1) = 0$

$n = 1$ $\quad 2n+1 = 3$

$H_3(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$

$H_3'(x) = a_1 + 2a_2 x + 3a_3 x^2$

$H_3(0) = f(0) = 0 \Rightarrow a_0 = 0$

$H_3(\frac{1}{2}) = f(\frac{1}{2}) = 1 \rightarrow a_0 + \frac{a_1}{2} + \frac{a_2}{4} + \frac{a_3}{8} = 1$

$H_3'(0) = f'(0) = \pi \Rightarrow a_1 = \pi$

$H_3'(\frac{1}{2}) = 0 \Rightarrow a_1 + 2a_2 \frac{1}{2} + 3a_3 \frac{1}{4} = 0$

$H_3(x) = \pi x + (12 - 4\pi) x^2 + (4\pi - 10) x^3$

$f(\frac{1}{4})$ tahmini Hatası

$f(x) = H_3(x) + R(x)$

$$R(x) = \frac{f^{(4)}(\xi)}{4!}(x-0)^2(x-\frac{1}{2})^2 \quad \xi \in (0,\frac{1}{2})$$

$$f(\frac{1}{4}) \simeq |H_3(\frac{1}{4})| = \frac{\pi}{4}(12-\pi)\frac{1}{16} + \frac{\pi^4}{24}|4\cdot\frac{1}{16} - 1|\frac{1}{64} \simeq 0,6367$$

$f^{(4)}(x) = \pi^4 \sin(\pi x)$

$$R(x) = \frac{\pi^4}{24}\sin(\pi x)\, x^2(x-\frac{1}{2})^2 \Big|_{x=1/4} = \frac{\pi^4}{24}\sin(\frac{\pi}{4})\frac{1}{4^2}\cdot\frac{1}{4^2}$$

$$|R(\tfrac{1}{4})| \leq \frac{\pi^4}{24\cdot 16\cdot 16} \simeq 0,0159$$

$$0,6208 \leq \sin(\tfrac{1}{4}) \leq 0,7122$$
$$0,7071$$

örnek2

x	$f(x)$	$f'(x)$
$x_0 = 0$	0,6	-0,5
$x_1 = 5$	0,4	-0,6
$x_2 = 1$	0,2	-1

$$L_{2,k}(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = 2x^2-3x+1$$

$$= -4x^2+4x$$
$$= 2x^2-x$$

$H_3(x) = ?$

$f(1,5)$ tahmin edin

$$\frac{L'_{2,k}(x)}{4x-3}$$

$$-8x+4$$

$$4x-1$$

$$H_{3x}(x)$$

$$[1-2(x-x_0)L'_{2,0}(x_0)]\cdot L^2_{2,0}(x) \quad \frac{H_{2,n}(x)}{(6x+1)(2x^2-3x+1)^2}\cdot f(0)$$
$$[1-2(x-x_1)L'_{2,1}(x_1)]\cdot L^2_{2,1}(x) \quad +(-4x^2+4x)^2 \cdot f(0,5)$$
$$[1-2(x-x_2)L'_{2,2}(x_2)]\cdot L^2_{2,2}(x) \quad +(-6x+7)(2x^2-x)^2 \cdot f(1)$$

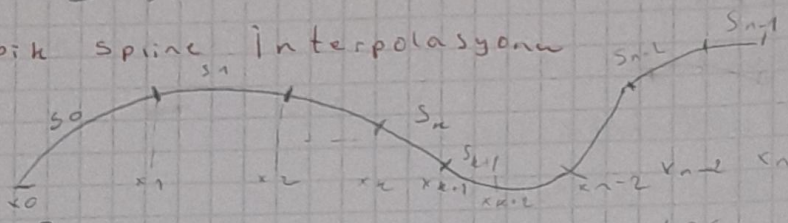
$$\hat{H}_{2,n}(x)$$

$$+(2x^2-3x+1)^2 \cdot f'(0)$$
$$+(x-0,5)(-4x^2+4x)^2 f'(0,5)$$
$$+(x-1)(2x^2-x)^2 f'(1)$$

$$H_3(x) = -6x^5+2x^4-1,2x^3+0,2x^2-0,5x-0,6$$

$f(1,5) \cong H_5(1,5) = -6,65$

kübik spline interpolasyonu



$[a,b]$ aralığında $f(x)$ fonksiyonu var

$a = x_0 < x_1 < \cdots < x_n = b$

$f(x_k)$, $k = 0, 1, \cdots, n$

$$S(x) \begin{cases} S_0(x) & a = x_0 \le x \le x_1 \\ S_1(x) & x_1 \le x \le x_2 \\ S_2(x) & x_2 \le x \le x_3 \\ \\ S_{n-1}(x) & x_{n-1} \le x \le x_n = b \end{cases}$$

"kübik spline interpolasyonu"

$S_k(x) = a_k + b_k(x-x_k) + c_k(x-x_k)^2 + d_k(x-x_k)^3$, $k = 0, \cdots, n-1$

koşullar

a) $S_0(x_0) = f(x_0)$, $S_k(x_{k-1}) = f(x_{k-1}) = S_{k-1}(x_{k-1})$, $k = 0, 1, \cdots, n-2$

$S_{n-1}(x_n) = f(x_n)$

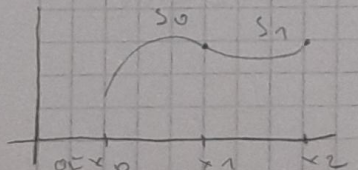
b) $S_k'(x_{k+1}) = S_{k+1}'(x_{k+1})$, $k = 0, 1, \cdots, n-2$

c) $S_k''(x_{k+1}) = S_{k+1}''(x_{k+1})$, $k = 0, 1, \cdots, n-2$

d) } uçlardan biri sağlanacak

(natural) doğal, serbest: 1) $S''(x_0) = 0 = S''(x_n) \Leftrightarrow S_0''(a) = S_{n-1}''(b) = 0$

bağlanmış, clamped: 2) $S'(x_0) = f'(x_0)$, $S'(x_n) = f'(x_n) \Leftrightarrow S_0'(a) = f'(a)$, $S_{n-1}'(b) = f'(b)$



$$S(x) = \begin{cases} S_0(x) & x_0 \le x \le x_1 \\ S_1(x) & x_1 \le x \le x_2 \end{cases}$$

$S_0(x) = a_0 + b_0(x-x_0) + c_0(x-x_0)^2 + d_0(x-x_0)^3$

$S_1(x) = a_1 + b_1(x-x_1) + c_1(x-x_1)^2 + d_1(x-x_1)^3$

a) $S_0(x_0) = f(x_0)$

$S_0(x_1) = f(x_1) = S_1(x_1)$

$S_1(x_2) = f(x_2)$

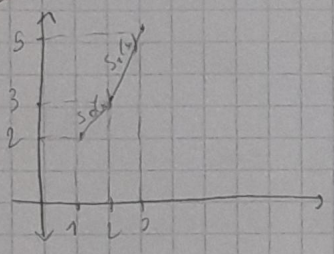
b) $S_0'(x_1) = S_1'(x_1)$

c) $S_0''(x_1) = S_1''(x_1)$

d) 1) $S_0''(x_0) = S_1''(x_2) = 0$

2) $S_0'(x_0) = f'(x_0)$, $S_1'(x_2) = f'(x_2)$

Ένα ακόμα



$x_0 = 1$ $f(x_0) = 2$

$x_1 = 2$ $f(x_1) = 3$

$x_2 = 3$ $f(x_2) = 5$

a) $S_0(1) = f(1) = 2 \Rightarrow a_0 = 2$

$S_0(2) = f(2) \Rightarrow a_0 + b_0 + c_0 + d_0 = 3$

$S_1(2) = f(2) \Rightarrow a_1 = 3$

$S_1(3) = f(3) \Rightarrow a_1 + b_1 + c_1 + d_1 = 5$

b) $S_0'(2) = S_1'(2) \Rightarrow b_0 + 2c_0 + 3d_0 = b_1$

c) $S_0''(2) = S_1''(2) \Rightarrow 2c_0 + 6d_0 = 2c_1$

"Dog-al" → d) $S_0''(1) = 0 \Rightarrow 2c_0 = 0$

$S_1''(2) = 0 \Rightarrow 2c_1 + 6d_1 = 0$

$$S(x) = \begin{cases} 2 + \frac{3}{4}(x-1) + \frac{1}{4}(x-1)^3, & 1 \le x \le 2 \\ 3 + \frac{3}{2}(x-2) + \frac{3}{4}(x-2)^2 - \frac{1}{4}(x-2)^3 & \end{cases}$$

$2 \le x \le 3$

Örnek4 : $[1,3]$ aralığında verilmiş $f(x)$ bağlamış koşullu kübik spline'ı,

$$S(x) = \begin{cases} S_0(x) = 3(x-1) + 2(x-1)^2 - (x-1)^3 & 1 \le x \le 2 \\ S_1(x) = a + b(x-2) + c(x-2)^2 + d(x-2)^3 & 2 \le x \le 3 \end{cases}$$

$f'(1) = f'(3)$ olursa a, b, c ve d ? cevap: $a=4$, $b=4$, $c=-1$, $d=\frac{1}{3}$